

K-NN algorithm

Lecture.5

KNN

K-Nearest Neighbor (KNN) is a simple and widely used machine learning technique for classification and regression tasks.

It works by identifying the K closest data points to a given input and making predictions based on the majority class or average value of those neighbors.

It is nonparametric method for classification

How to works K-NN

- Choose the number of neighbors k .
- Compute the distance between the new point and all points in the training set. Common distance metrics:
 - Euclidean distance (most common)
 - Manhattan distance
 - Minkowski distance “
- Select the k nearest neighbors.
- For **classification**: choose the class that is most frequent among neighbors.
For **regression**: take the average of neighbors' values

- Small k: more sensitive to noise → **may overfit**.
- Large k : smoother decision boundary → **may underfit**.
- Common practice: try several values of k and pick the one with best cross-validation performance
- **Euclidean distance** is defined as the straight-line distance between two points in a plane or space. You can think of it like the shortest path you would walk if you were to go directly from one point to another.

$$\text{distance}(x, X_i) = \sqrt{\sum_{j=1}^d (x_j - X_{i_j})^2}$$

Age	Loan	Default	Distance	
25	\$40,000	N	102000	
35	\$60,000	N	82000	
45	\$80,000	N	62000	
20	\$20,000	N	122000	
35	\$120,000	N	22000	2
52	\$18,000	N	124000	
23	\$95,000	Y	47000	
40	\$62,000	Y	80000	
60	\$100,000	Y	42000	3
48	\$220,000	Y	78000	
33	\$150,000	Y	8000	1
48	\$142,000	?		

Euclidean Distance

$$D = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

Name	Aptitude	Communication	Class
Karuna	2	5	Speaker
Bhuvna	2	6	Speaker
Gaurav	7	6	Leader
Parul	7	2.5	Intel
Dinesh	8	6	Leader
Jani	4	7	Speaker
Bobby	5	3	Intel
Parimal	3	5.5	Speaker
Govind	8	3	Intel
Susant	6	5.5	Leader
Gouri	6	4	Intel
Bharat	6	7	Leader
Ravi	6	2	Intel
Pradeep	9	7	Leader
★ Josh	5	4.5	???

Name	Aptitude	Communication	Class	Distance	$k = 1$	$k = 2$	$k = 3$
Karuna	2	5	Speaker	3.041			
Bhuvna	2	6	Speaker	3.354			
Parimal	3	5.5	Speaker	2.236			
Jani	4	7	Speaker	2.693			
Bobby	5	3	Intel	1.500			1.500
Ravi	6	2	Intel	2.693			
Gouri	6	4	Intel	1.118	1.118	1.118	1.118
Parul	7	2.5	Intel	2.828			
Govind	8	3	Intel	3.354			
Susant	6	5.5	Leader	1.414			
Bharat	6	7	Leader	2.693			
Gaurav	7	6	Leader	2.500			
Dinesh	8	6	Leader	3.354			
Pradeep	9	7	Leader	4.717			
Josh	5	4.5	???				

Problem KNN

1. When variables have different measurement scales: When calculating Euclidean distance, the **income difference** will be much larger numerically than the **age difference**, so income dominates the distance, even if age is actually important.
2. there is a mixture of numerical and categorical variables.

To solve this,

- **standardize features** so they have the **same scale**:

$$x_{\text{standardized}} = \frac{x - \mu}{\sigma}$$

μ = mean of the feature

σ = standard deviation of the feature

- **Min-Max is scaling features to [0,1]:**

$$x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Example 2:

Consider a dataset that contains two variables: height (cm) & weight (kg). Each point is classified as normal or underweight.

Weight(x2)	Height(y2)	Class
51	167	Underweight
62	182	Normal
69	176	Normal
64	173	Normal
65	172	Normal
56	174	Underweight
58	169	Normal
57	173	Normal
55	170	Normal

The following table shows the calculated Euclidean distance of unknown data points from all points.

Weight(x2)	Height(y2)	Class	Euclidean Distance
51	167	Underweight	6.7
62	182	Normal	13
69	176	Normal	13.4
64	173	Normal	7.6
65	172	Normal	8.2
56	174	Underweight	4.1
58	169	Normal	1.4
57	173	Normal	3
55	170	Normal	2

Now, we have a new data point (x1, y1), and we need to determine its class.

Weight(x2)	Height(y2)	Class	Euclidean Distance
51	167	Underweight	6.7
62	182	Normal	13
69	176	Normal	13.4
64	173	Normal	7.6
65	172	Normal	8.2
56	174	Underweight	4.1
58	169	Normal	1.4
57	173	Normal	3
55	170	Normal	2



Since the majority of neighbors are classified as normal as per the KNN algorithm, the data point (57, 170) should be normal.

DT revise

- If the **information gain is 0**, it means that there is no reduction in entropy due to split of the data set according to that particular feature.

DT revise

CGPA	Communication	Aptitude	Programming Skill	Job offered?
High	Good	High	Good	Yes
Medium	Good	High	Good	Yes
Low	Bad	Low	Good	No
Low	Good	Low	Bad	No
High	Good	High	Bad	Yes
High	Good	High	Good	Yes
Medium	Bad	Low	Bad	No
Medium	Bad	Low	Good	No
High	Bad	High	Good	Yes
Medium	Good	High	Good	Yes
Low	Bad	High	Bad	No
Low	Bad	High	Bad	No
Medium	Good	High	Bad	Yes
Low	Good	Low	Good	No
High	Bad	Low	Bad	No
Medium	Bad	High	Good	No
High	Bad	Low	Bad	No
Medium	Good	High	Bad	Yes

$$\text{Entropy}(S) = -0.44 \log_2(0.44) - 0.56 \log_2(0.56) = 0.99.$$

Information gain of CGPA=0.30

Information gain of communication=0.36

Information gain of aptitude=0.47

Information gain of programming skills=0.04

Second split from high aptitude

0.85 before the split

0.52 when the feature 'CGPA' is used for split

0.55 when the feature 'Communication' is used for split

0.05 when the feature 'Programming skill' is used for split

Third split from bad communication

0.81 before the split

0 when the feature 'CGPA' is used for split

0.50 when the feature 'Programming Skill' is used for split